Software Requirements Specification (SRS)

Smart Fleet Management System (Smart FM)

Client: ABC-Trans Logistics
Unit: SWE30003 - Software Architectures and Design
University: Swinburne University of Technology
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1. Project Goals, Assumptions, and Context

Project Type: Enterprise Information System / Fleet Management Information System

SmartFM is a centralized enterprise information system designed to support logistics and transportation operations across multiple branches. The system combines customer-facing services (online booking, payment, shipment tracking) with internal operational services (fleet management, driver allocation, reporting, and dispatching).

1.1 Project Context and Pain Points

ABC-Trans is a rapidly expanding logistics transportation company in Vietnam. Currently, operations rely on disconnected, branch-specific systems and manual processes. This fragmentation results in significant pain points for the business:

- Customer Dissatisfaction: Slow service, unavailability of transparent tracking, and cumbersome ordering processes.
- Operational Inefficiencies: Branch managers struggle with manual vehicle assignments, leading to resource underutilization and delays.
- Data Fragmentation: Lack of centralized statistics prevents management from making informed, data-driven decisions regarding fleet expansion and branch performance.

1.2 Business Goals

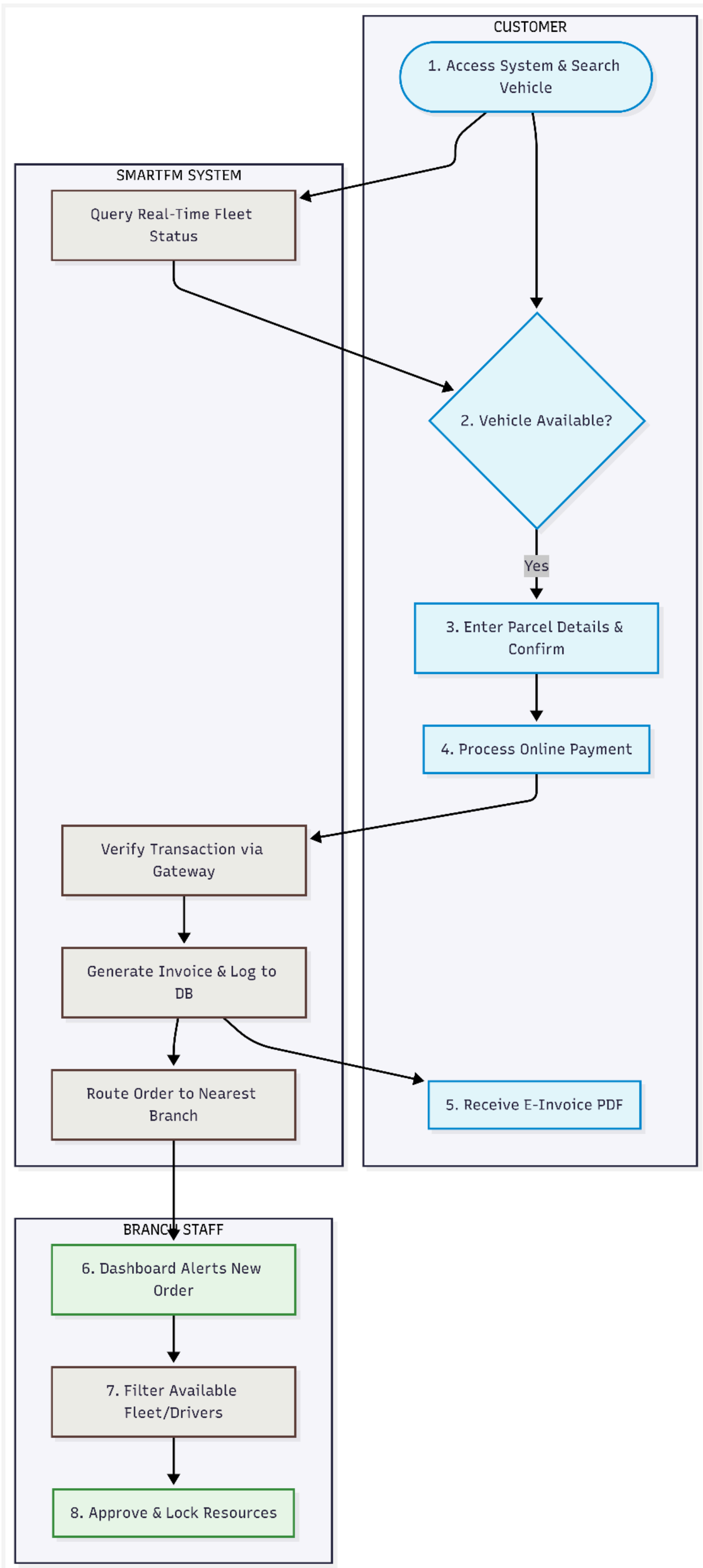
The SmartFM system addresses several key business goals:

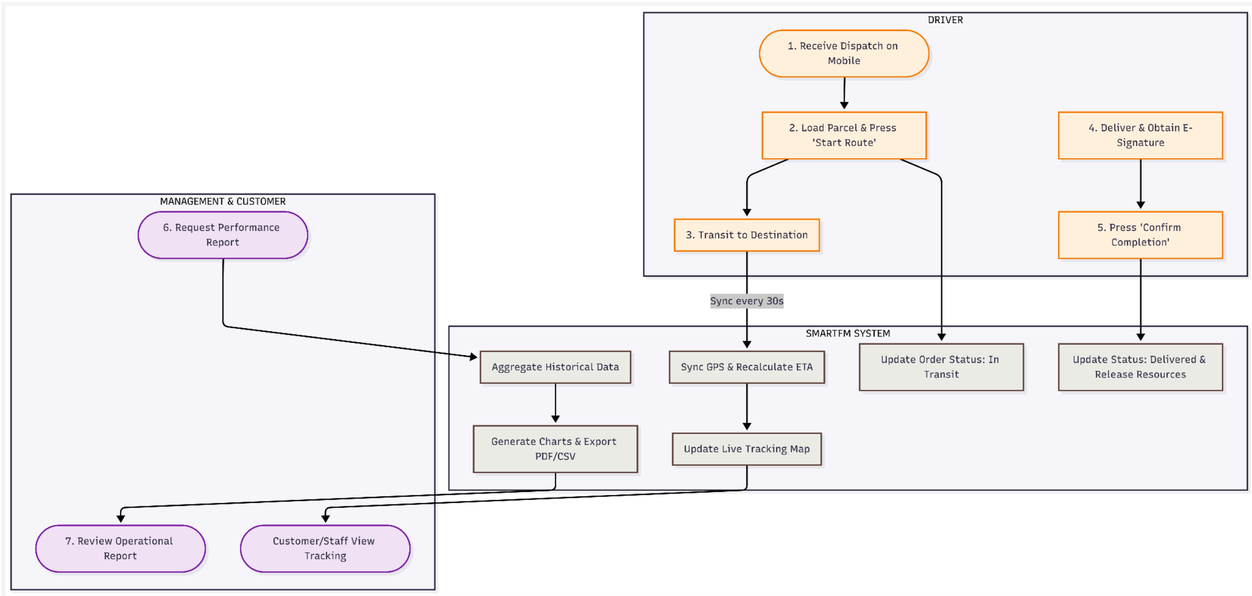
- Reduce Cost of Operations: Automate manual administrative tasks like invoicing, order logging, and assignment to reduce labor overhead and minimize routing delays.
- Improve Quality of Service: Provide customers with a seamless, 24/7 online ordering platform and real-time shipment visibility.
- Improve Capabilities: Enable multi-platform access and integrate digital payment gateways, transitioning from a cash-heavy, branch-dependent model.
- Improve Market Position: Modernize ABC-Trans's technological footprint to compete with leading tech-driven logistics providers in Vietnam.

1.3 Assumptions and Constraints

- Infrastructure: All ABC-Trans branches have stable internet access to interact with the centralized cloud database.
- External Services: Third-party APIs for mapping/GPS for real-time tracking and secure payment gateways for credit/debit processing are available and reliable.
- Scope Limitation: As explicitly stated by the client, advanced features such as loyalty programs and dynamic discount modules are deferred to future iterations and are strictly out of scope for this initial SRS.

2. High-Level Workflow



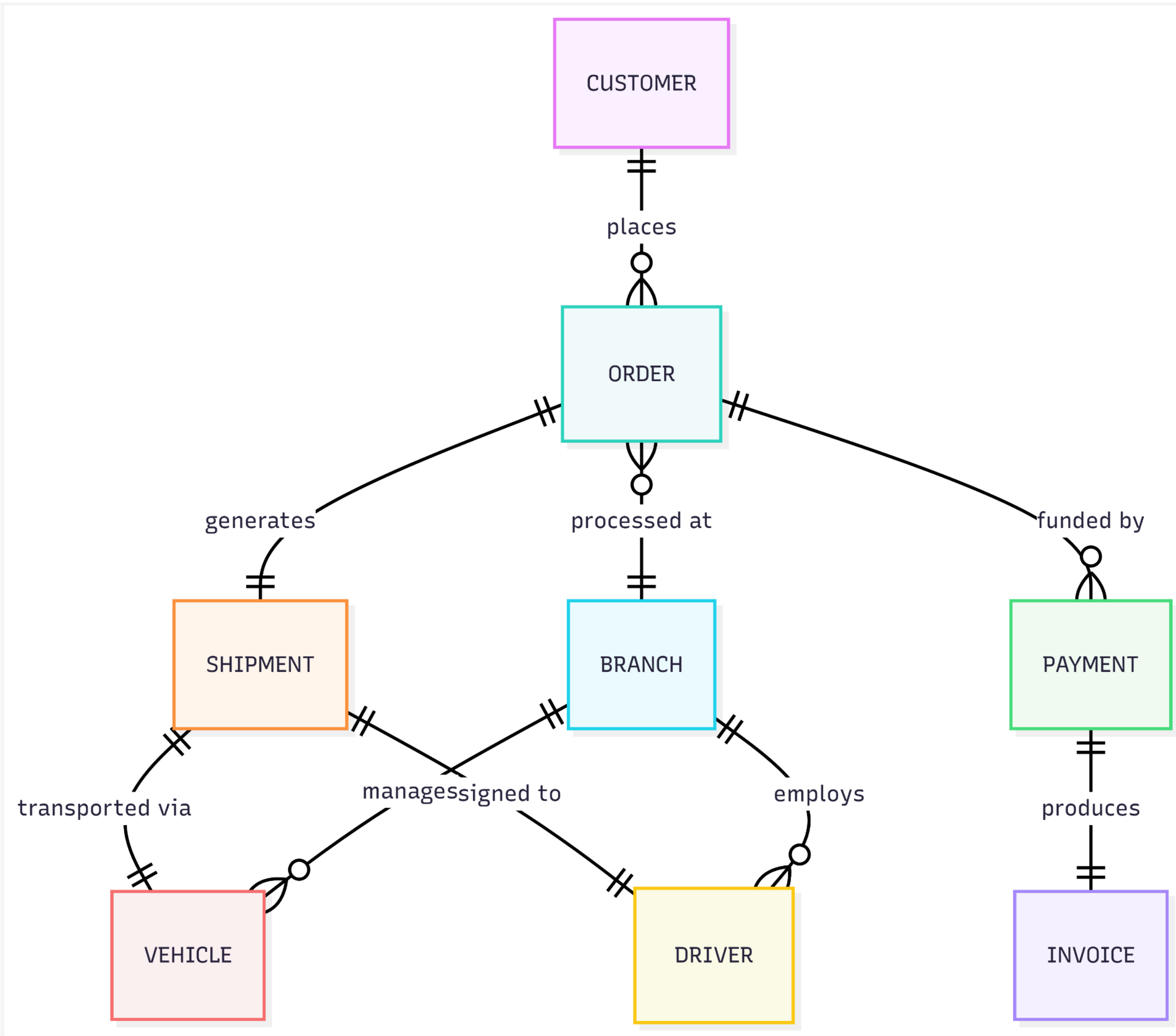


Workflow Presentation and Diagram Notation Explanation:

The activity diagrams above illustrate the core workflows across different users and the system. The rounded rectangles signify specific procedural steps, arrows indicate the sequential logic control flow, and diamonds represent decision branch points.

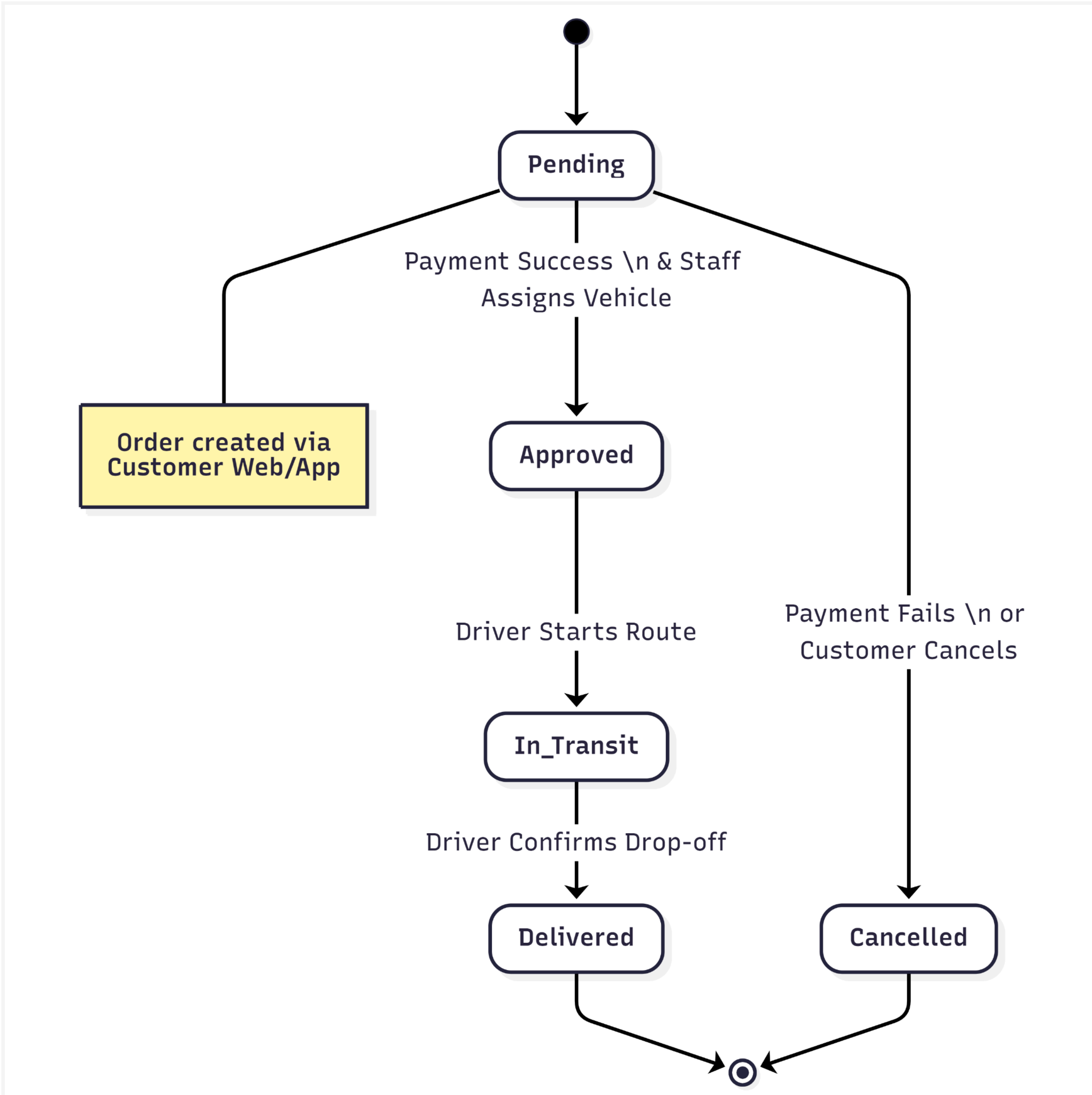
- Workflow 1 (Order Placement and Resource Allocation): Integrates Tasks 1, 2, 3, and 4. It ensures that a customer cannot book an unavailable asset and locks the physical truck and driver immediately upon successful payment verification to prevent double-booking issues across branches.
- Workflow 2 (Shipment Delivery and Real-Time Tracking): Integrates Tasks 5, 6, 7, and 8. It highlights the fully automated synchronization between the Driver's mobile GPS hardware data and the centralized SmartFM back-end cloud database, eliminating the need for manual check-ins and providing customers with live updates.

3. Domain / Data Model



3.1 Entities Descriptions

Source Entity	Relationship	Target Entity	Description / Multiplicity
Customer	places	Order	One Customer can place many Orders (1..N).
Order	processed at	Branch	Many Orders are handled by one Branch (N..1).
Order	generates	Shipment	One Order results in one physical Shipment (1..1).
Order	funded by	Payment	One Order can have one or multiple Payments (1..N).
Payment	produces	Invoice	One Payment generates one Invoice receipt (1..1).
Shipment	transported via	Vehicle	One Shipment is assigned to one Vehicle at a time (1..1).
Shipment	assigned to	Driver	One Shipment is handled by one Driver (1..1).
Branch	manages	Vehicle	One Branch manages many Vehicles (1..N).
Branch	employs	Driver	One Branch employs many Drivers (1..N).



Order Lifecycle :

The diagram below maps the order lifecycle from creation to completion. It highlights how actions such as successful payments or driver drop-off confirmations shift an order from Pending to either Delivered or Cancelled. Properly managing these states is key to keeping our data consistent across the system.

4. Functional Requirements: Task and Support Descriptions

Task 1: Search and Verify Availability

Task:	1.1 Search Fleet and Route Availability		
Purpose:	Allow customers to find available vehicles and shipment slots for specific locations and dates before booking.		
Users:	Customers (Web/Mobile App)	Frequency:	High (Hundreds per day across branches)
Trigger / Precondition:	Customers access the platform with the intent to ship goods.		
Sub-tasks		Example Solutions	
1. Specify pickup/drop-off locations and desired dates. Problem: Customer does not know which branch covers their area.		The system provides an interactive map and predictive text fields to automatically map addresses to the nearest ABC-Trans branch.	
2. View available vehicles/slots. Problem: Overbooking or requesting vehicles that are under maintenance.		System queries the centralized database in real-time and exclusively displays vehicle classes currently marked "Available" for the selected dates.	
3. Compare pricing and transit times.		The system dynamically calculates and displays a comparison grid of estimated costs and ETAs based on distance and vehicle type.	
Variants:	1a. No vehicles available at the nearest branch -> System suggests next available date or an adjacent branch.		

Task 2: Place Shipment Order

Task:	1.2 Book Shipment		
Purpose:	Officially record the customer's request for logistics service into the system.		
Users:	Customers on Web/Mobile or Staff	Frequency:	High
Trigger / Precondition:	The customer has selected a valid vehicle and time slot from Task 1.		
Sub-tasks		Example Solutions	
1. Enter sender, receiver, and parcel specifications. Problem: Incomplete or invalid contact information leads to failed deliveries.		The system provides structured web forms with strict data validation (regex for phone numbers, mandatory weight/dimension fields).	
2. Confirm order details. Problem: Customers mistype addresses.		The system generates a comprehensive Order Summary screen for final review before committing to the database.	
Variants:	2a. The customer is a registered user -> System auto-fills sender details. 2b. Order placed in-branch -> Staff enters data on behalf of the customer.		

Task 3: Make Payment and Receive Invoice

Task:	1.3 Process Payment and Invoice		
Purpose:	Securely finalize the financial transaction and generate legal records.		
Users:	Customers, Branch Cashiers	Critical:	High (Financial Data)
Trigger / Precondition:	Order details are confirmed and the final price is calculated.		

Sub-tasks	Example Solutions
1. Select payment method.	The system presents options: Credit/Debit Card, Digital Wallet, or Pay-at-Branch.
2. Execute transaction. Problem: Credit card fraud or failed transactions.	System redirects to a secure third-party payment gateway; handles success/failure callbacks gracefully.
3. Obtain receipt. Problem: Paper receipts get lost.	The system automatically generates a PDF invoice, emails it to the customer, and logs it in the central database.
Variants:	3a. Payment fails -> System prompts user to try another method. 3b. Cash payment in-branch -> Staff manually marks order as Paid in the system.

Task 4: Process Incoming Orders and Approve Assignments

Task:	2.1 Order Processing and Resource Allocation		
Purpose:	Allow branch staff to review customer orders and firmly allocate physical vehicles and drivers.		
Users:	Branch Staff / Operations	Frequency:	Continuous during operational hours
Trigger / Precondition:	A new order enters the system with Paid or Pending Payment status.		
Sub-tasks		Example Solutions	
1. Identify unassigned orders. Problem: Orders get missed in high-volume periods.		The system provides a unified dashboard highlighting new, unassigned orders with SLA countdown timers.	
2. Assign a specific vehicle and driver. Problem: Assigning a driver who is already		The system cross-references the roster and fleet database, only allowing selection from a	

on shift or a full truck.	dropdown of currently available resources.
3. Approve the order.	System updates the Order Status to Approved and locks the assigned resources.
Variants:	4a. Resources unexpectedly unavailable -> Staff places order on Hold and system notifies customer of delay.

Task 5: Organize Delivery and Pickup

Task:	2.2 Dispatching		
Purpose:	Coordinate the physical departure of the fleet and notify drivers of their routes.		
Users:	Dispatchers, Drivers	Critical:	Medium
Trigger / Precondition:	The order is approved, and the scheduled dispatch time approaches.		
Sub-tasks		Example Solutions	
1. Inform the driver of the route. Problem: Paper manifests get lost or misread.		The system pushes a digital manifest and optimized route directly to the Driver's mobile application.	
2. Initiate the physical transit.		The driver presses Start Route on the app, updating the system status to In Transit.	

Task 6: Track Shipment in Real-Time

Task:	3.1 Shipment Tracking		
Purpose:	Provide visibility over the shipment's geographical location and expected delivery time.		
Users:	Customers, Branch Staff	Frequency:	Very High

Trigger / Precondition:	Shipment is marked as In Transit.		
Sub-tasks		Example Solutions	
1. Query shipment status. Problem: Customers overwhelm call centers asking, "Where is my package?"		The system provides a public tracking portal via Web/App where users can input their Order ID.	
2. View current location and ETA. Problem: Static tracking only shows "Dispatched".		The system integrates with the Driver's mobile GPS to update coordinates to a map API, displaying visual location and dynamic ETAs.	

Task 7: Manage Vehicle and Driver Information

Task:	4.1 Resource Maintenance		
Purpose:	Keep the central database accurate regarding the status of the fleet and personnel.		
Users:	Branch Managers, System Admins	Critical:	High (Data Integrity)
Trigger / Precondition:	A new vehicle is purchased, a driver is hired, or an asset requires maintenance.		
Sub-tasks		Example Solutions	
1. Update asset status. Problem: A branch tries to book a truck that is currently in the repair shop.		The system provides a secure CRUD interface for Managers to toggle vehicle statuses, for example: Active or In Maintenance	
2. Ensure global data consistency.		The system immediately synchronizes updates to the central database so the online booking portal reflects accurate capacity.	

Task 8: Generate Statistics and Utilization Reports

Task:	5.1 Managerial Reporting		
Purpose:	Extract business intelligence from operational data to support decision-making.		
Users:	Upper Management	Frequency:	Weekly/Monthly
Trigger / Precondition:	Management initiates an audit or periodic review.		
Sub-tasks (User Intentions / Domain Problems)		System Support (Example Solutions)	
1. Define reporting parameters. Problem: Manual compilation of data from different branch systems takes days.		The system provides a Reporting Module with filters for Date Range in Day, Week, Month, YTD, Branch, and Asset Type.	
2. Analyze data and export.		The system aggregates the central data into visual charts and allows export to CSV/PDF formats.	

5. Quality Attributes (Non-Functional Requirements)

The system must adhere to the following key quality attributes, defined using verifiable Quality Attribute Scenarios encompassing Stimulus, Response, and Measure.

5.1 Performance

- Stimulus: A customer initiates a search for vehicle availability during peak business hours.
- Response: The system queries the centralized database across all branches and returns the matching available vehicles.
- Measure: The search results must be displayed to the user within 2.0 seconds for 95% of queries.
- Rationale: High latency during the booking process leads directly to shopping cart abandonment and customer dissatisfaction.

5.2 Availability

- Stimulus: A hardware fault occurs on the primary web server hosting the online booking portal.
- Response: The system automatically fails over to a secondary redundant server to maintain operational continuity.
- Measure: The system must maintain an overall uptime of 99.9% per month, resulting in no more than approximately 43 minutes of downtime monthly.
- Rationale: As branch dispatching and customer orders rely entirely on this centralized platform, prolonged downtime halts ABC-Trans's physical business operations.

5.3 Security

- Stimulus: A malicious actor attempts to intercept network traffic between the customer's mobile app and the payment gateway.
- Response: The system encrypts all data in transit and rejects any unauthorized connection attempts.
- Measure: 100% of customer payment data must be encrypted using AES-256 or higher, and the system must pass PCI-DSS compliance audits, never storing CVV numbers locally.
- Rationale: Financial and personal data must be protected to prevent legal liabilities and maintain customer trust.

5.4 Usability

- Stimulus: A new customer with basic digital literacy attempts to place a shipment order on the web platform for the first time.
- Response: The system guides the user through the booking process using intuitive UI patterns and clear form validations.
- Measure: A first-time user must be able to complete the Place Shipment Order task from start to payment within 3 minutes, without needing to consult a help manual or customer support.
- Rationale: A steep learning curve will deter customers from utilizing the new self-service digital platforms, defeating the modernization goal.

6. Other Requirements (Product and Design Level)

- Multi-Platform Accessibility: The client-facing product must be deployed as a responsive Web Application compatible with modern browsers such as Chrome, Edge, Safari,... and as a native or cross-platform Mobile Application like IOS and Android.
- Centralized Data Architecture: The design must utilize a single, centralized database or logically centralized distributed database to ensure that vehicle status, driver rosters, and order states are globally consistent across all ABC-Trans branches in real-time.
- Modularity for Future Expansion: The architecture must be loosely coupled, allowing the

future integration of Loyalty Program and Discount Modules without requiring a rewrite of the core ordering and payment logic.

- UI/UX Guidelines: The User Interface must conform strictly to the internal Swinsoft Consulting UI/UX Corporate Standards.

7. Requirements Validation

To ensure the requirements accurately reflect ABC-Trans's needs and trace back to the business goals, the following validation activities have been designed:

1. Stakeholder Task Walkthroughs: We will conduct line-by-line reviews of the 8 Task and Support descriptions with representative ABC-Trans Branch Managers and Dispatchers to confirm that the identified domain problems match their day-to-day physical realities, and that the proposed system support solves those issues.
2. Goal-Domain Tracing: We have cross-referenced the Domain Model and User Tasks against the initial business goals. For example, Task 8 directly supports the goal of improving statistics and resource utilization.
3. UI Prototyping: Low-fidelity wireframes of the online booking process will be presented to sample customers to validate the workflow before heavy development begins.

CRUD Check

Task / Entity	Customer	Order	Shipment	Vehicle	Driver	Pay ment	Invoice	Branch
Search Availability	R			R				R
Book Shipment	C, R	C, R						
Payment and Invoice	R	R				C, U	C	
Resource Allocation	R	U, R	C, U	R, U	R, U			R
Dispatching		R	U	R	R			
Tracking	R	R	R, U	R	R			
Resource Management				C, R, U	C, R, U			R
Reporting	R	R	R	R	R	R	R	R

8. Requirements Verifiability

The requirements specified in this document have been reviewed to ensure that they are verifiable. To improve verifiability, requirements have been written using measurable criteria wherever possible.

The following table demonstrates how key requirements can be verified:

Requirement	Verification Method
Vehicle availability search results shall be displayed within 2 seconds for 95% of requests.	Performance testing under simulated peak load conditions.
The system shall maintain 99.9% monthly availability.	Operational monitoring and uptime reports.
Customers shall be able to place shipment orders through web and mobile platforms.	Functional testing of order creation workflows on supported platforms.
The system shall generate an invoice automatically following successful payment.	Functional testing of payment and invoicing processes.
Customers shall be able to track shipments in real time.	Integration testing with GPS tracking services and acceptance testing with representative users.
Vehicle and driver information shall be synchronized across all branches.	Database consistency testing and multi-branch scenario testing.
The system shall support multiple payment methods, including card, digital wallet, and cash payment.	Functional testing of each payment workflow.
Management shall be able to generate shipment and utilisation reports.	Functional testing of reporting features using sample operational data.
A first-time customer shall be able to complete a shipment booking within 3 minutes.	Usability testing with representative users performing the booking task.
The system shall export reports in PDF and CSV formats.	Functional testing of report generation and export features.

9. Alternative Solutions

Solution 1: Cloud-Based Centralized SmartFM Platform

A cloud-based centralized platform is the most suitable solution for ABC-Trans because it allows all branches to access the same real-time data regarding vehicles, drivers, shipments, payments, and customer orders. Customers can interact with the system through web and mobile applications, while branch staff can manage dispatching, resource allocation, and reporting through a unified interface. This approach directly addresses the current problems of fragmented branch systems and inconsistent data by ensuring that all operational information is synchronized across the organization.

This approach is widely adopted within the logistics industry. For example, global logistics companies such as DHL have implemented cloud-based fleet management technologies to coordinate large vehicle fleets and provide real-time operational visibility. Similarly, modern fleet management platforms have demonstrated significant improvements in delivery efficiency, fleet utilization, and operational control by integrating vehicle tracking, driver management, routing, and invoicing into a single centralized environment. These examples demonstrate that a centralized cloud architecture is a proven and scalable solution for logistics organizations operating across multiple locations.

The primary advantages of this solution include real-time information sharing, improved scalability as ABC-Trans expands, reduced maintenance costs through centralized administration, and improved customer experience through online ordering and shipment tracking. A potential disadvantage is the reliance on stable internet connectivity across all branches, although this assumption has already been identified within the project constraints.

Solution 2: Intelligent Route Optimization and Dispatching System

A second solution approach focuses on integrating advanced route optimization and automated dispatching capabilities into the SmartFM platform. Rather than relying on manual planning by branch staff, the system would automatically assign vehicles and drivers, calculate efficient delivery routes, and continuously update schedules based on traffic conditions, delivery commitments, and vehicle availability. This solution directly addresses operational inefficiencies, resource underutilization, and delivery delays identified in the case study.

A well-known real-world example is UPS's ORION (On-Road Integrated Optimization and Navigation) system. ORION uses optimization algorithms and fleet data to generate efficient delivery routes for drivers and has helped UPS significantly reduce vehicle mileage and fuel consumption. UPS reports that route optimization technologies have enabled substantial reductions in travel distance and fuel usage across its delivery network. These results demonstrate the value of intelligent dispatching and route optimization in large-scale logistics operations.

The major benefits of this solution include lower fuel costs, improved fleet utilization, faster deliveries, and reduced workload for dispatch staff. Additionally, optimized routing supports future scalability by enabling the company to manage a larger number of deliveries without proportionally increasing operational resources. The primary disadvantage is the increased complexity of implementation and the need for accurate GPS, mapping, and vehicle-location data.